Campus Lost & Found Intelligence System

DBMS Experiential Learning – Level 3 Project Synopsis

1. Problem Statement

Campus lost-and-found processes are often informal and fragmented. Students may report lost or found items through notices, messaging groups, or word of mouth, making it difficult to discover potential matches and verify claims. This project proposes a centralized web application that records lost and found items and intelligently identifies possible matches using item descriptions, attributes, location, and time information.

2. Objectives

• Provide a centralized platform for students to report and search lost and found items.

• Maintain structured records of users, reports, locations, claims, and item status.

• Use MongoDB to store flexible item descriptions and attributes that vary across item categories.

• Identify and rank potential matches between lost and found reports.

• Provide an authenticated workflow for reporting, claiming, verification, and return of items.

3. Proposed System

The system will provide separate interfaces for Students and Administrators. Students can submit lost/found reports, search active reports, view potential matches, and submit claims. Administrators can verify reports and claims, manage item status, and monitor system activity. A matching component will compare relevant attributes such as category, brand, location, date/time, and textual descriptions to generate potential-match results.

4. Database Design

MySQL	MongoDB
Structured transactional data: Users, LostItems, FoundItems, Locations, Claims, and status/history information.	Flexible item data: descriptions, category-specific attributes, visual descriptions, additional details, and report metadata.

5. Key Database Features

The project will demonstrate DDL and DML operations, primary/foreign keys, constraints, indexes, joins, aggregation, views, transactions, stored procedures/functions, triggers, and MongoDB aggregation. Transactions will be used for claim approval and item-status updates, while triggers can maintain consistent status transitions.

6. Technology Stack

Frontend: Next.js **Backend:** Next.js API Routes/Route Handlers **Databases:** MySQL + MongoDB **Version Control:** Git/GitHub

7. Expected Outcome

The final system will provide a functional campus lost-and-found platform that reduces the difficulty of locating reported items and supports a controlled claim-and-return process. The project will demonstrate how relational and document databases can be combined according to the structure and requirements of the data, while going beyond basic CRUD functionality.